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members of the branch would possess the derived character which will not be seen in organisms below the particular branch point. Arranging organisms on the basis of their similar or derived characters which differ from the ancestral characters produced a phylogenetic tree or cladogram (Figure 1.1).

Depending on the system of classification, organisms were classified into two or three kingdoms. Later into four, five, six and now into seven kingdoms. R.H.Whittaker (1969) proposed the **Five kingdom Classification**, the Kingdoms defined by him were Monera, Protista, Fungi, Plantae, and Animalia based on the cell structure, mode of nutrition, mode of reproduction and phylogenetic relationships. Table 1. gives a comparative account of different characteristics of the five kingdoms.

Classification has come a long way and now takes into an account even molecular level DNA and RNA identification. The advancement in molecular techniques and biochemical assays has led to a new classification - The "Three Domain" classification.

1.4. Three Domains of life

Three domain classification was proposed by **Carl Woese** (1977) and his co-workers. They classified organisms based on the difference in 16S rRNA genes. The three domain system adds the taxon 'domain' higher than the kingdom. This system emphasizes the separation of Prokaryotes into two domains, Bacteria and Arachaea, and all the eukaryotes are placed into the domain Eukarya. Archaea appears to have more in common with the Eukarya than the Bacteria. Archaea differ from bacteria in cell wall composition and differs from bacteria and eukaryotes in membrane composition and rRNA types.



Thermus aquatics is a bacterium which can tolerate high temperatures. The first DNA polymerase enzyme was isolated from *T. aquaticus* it is used in **PCR** (Polymerase Chain Reaction) for DNA amplification.

